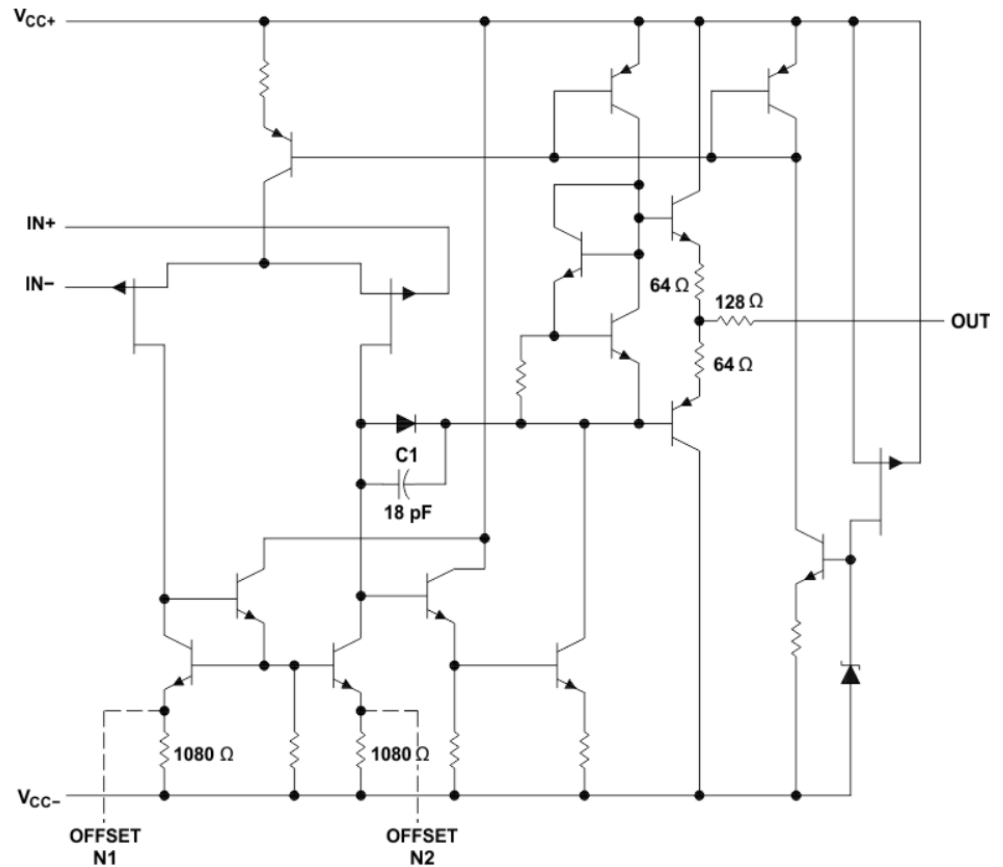
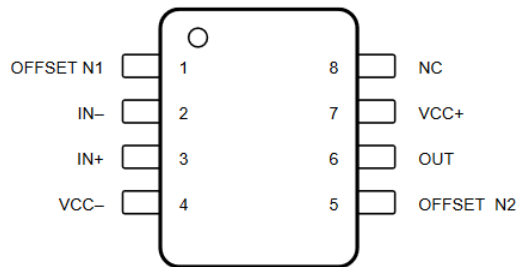
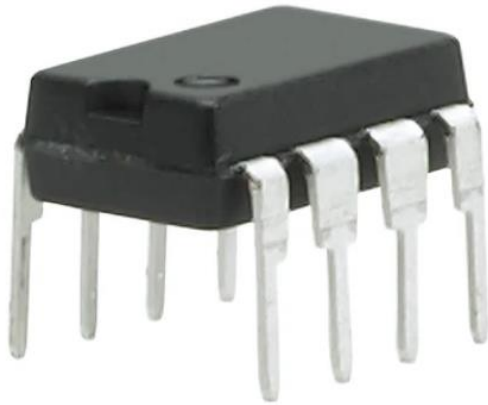
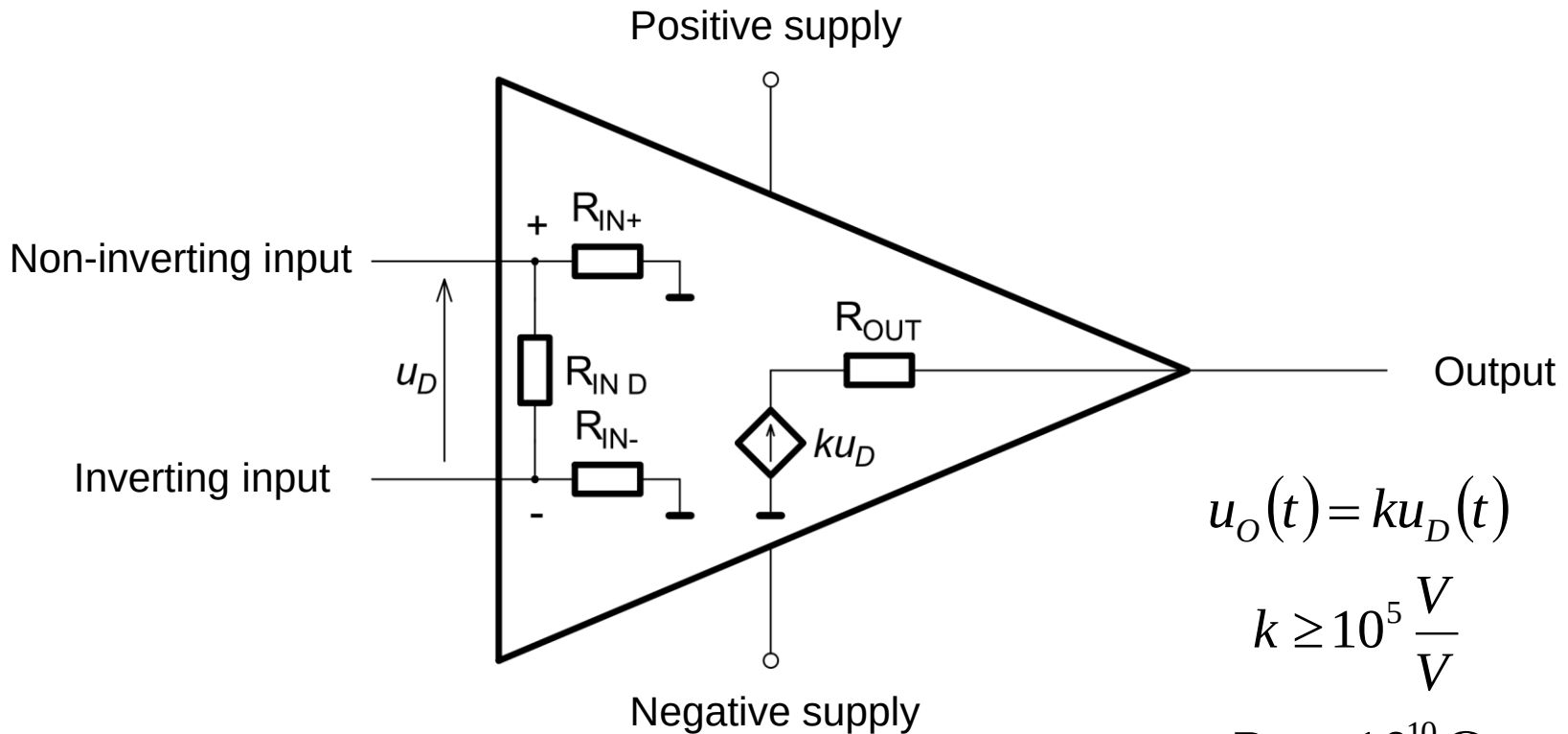


Introduction to Operational Amplifiers



Introduction to Operational Amplifiers

SIMPLIFIED SCHEMATIC DIAGRAM OF A REAL OPERATIONAL AMPLIFIER



$$u_O(t) = ku_D(t)$$

$$k \geq 10^5 \frac{V}{V}$$

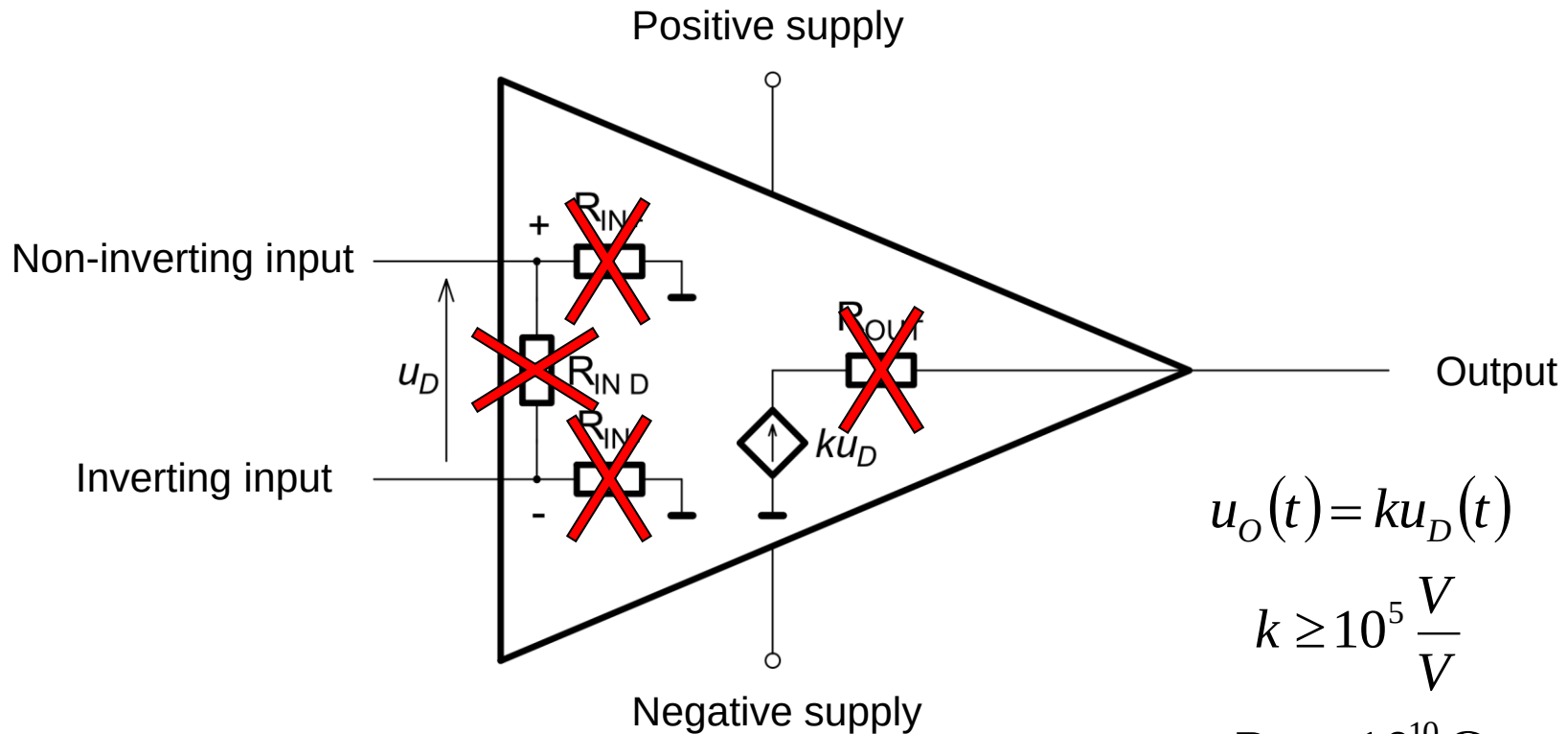
$$R_{IN} \geq 10^{10} \Omega$$

$$R_{OUT} < 10 \Omega$$

(supressed by
circuit's feedback)

Introduction to Operational Amplifiers

SCHEMATIC DIAGRAM OF AN IDEAL OPERATIONAL AMPLIFIER



$$u_O(t) = k u_D(t)$$

$$k \geq 10^5 \frac{V}{V}$$

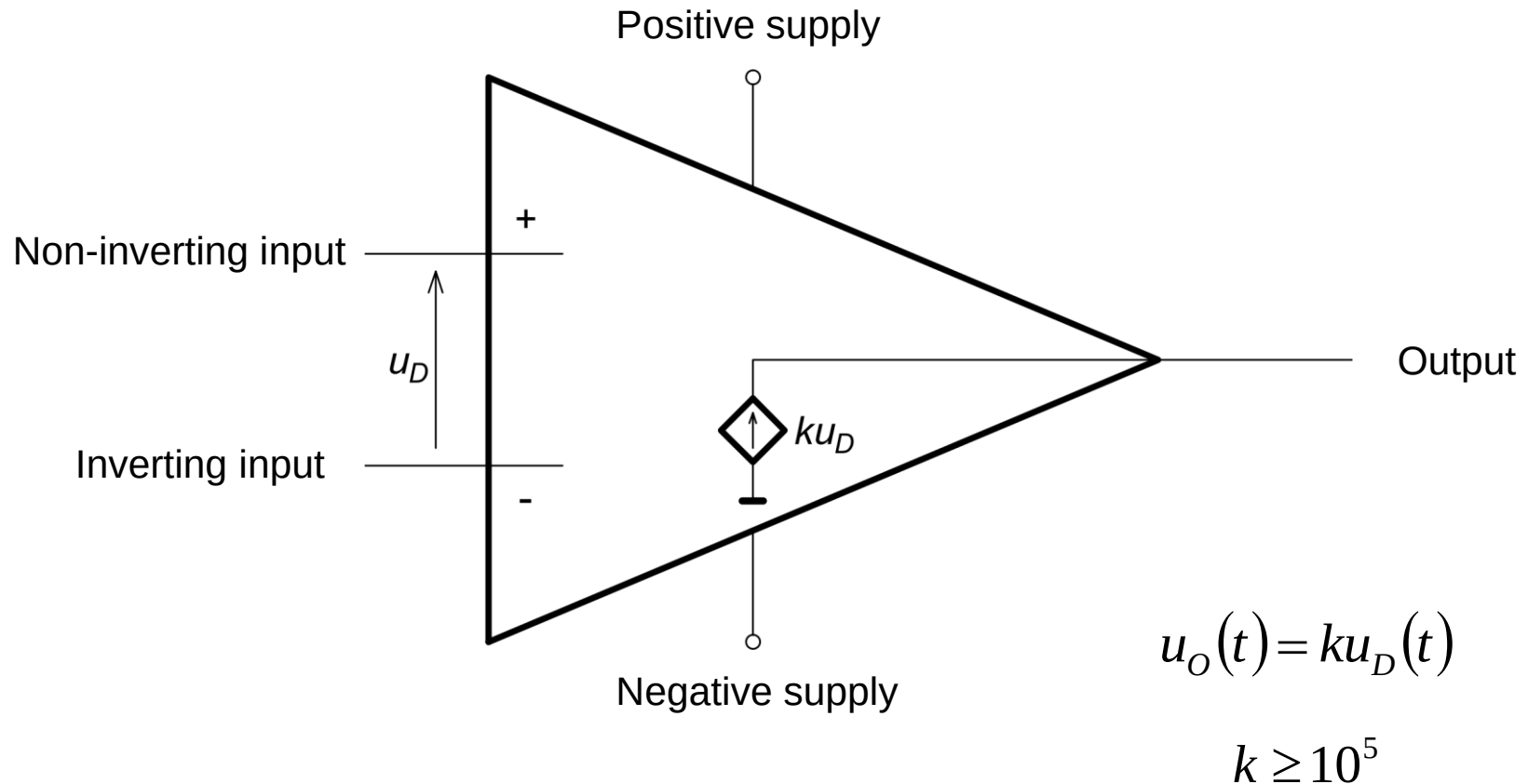
$$R_{IN} \geq 10^{10} \Omega$$

$$R_{OUT} < 10 \Omega$$

(supressed by
circuit's feedback)

Introduction to Operational Amplifiers

SCHEMATIC DIAGRAM OF AN IDEAL OPERATIONAL AMPLIFIER



Introduction to Operational Amplifiers

BASIC AMPLIFIER: NON-INVERTING CONFIGURATION WITH FINITE AMPLIFICATION k

$$u_O = k u_D$$

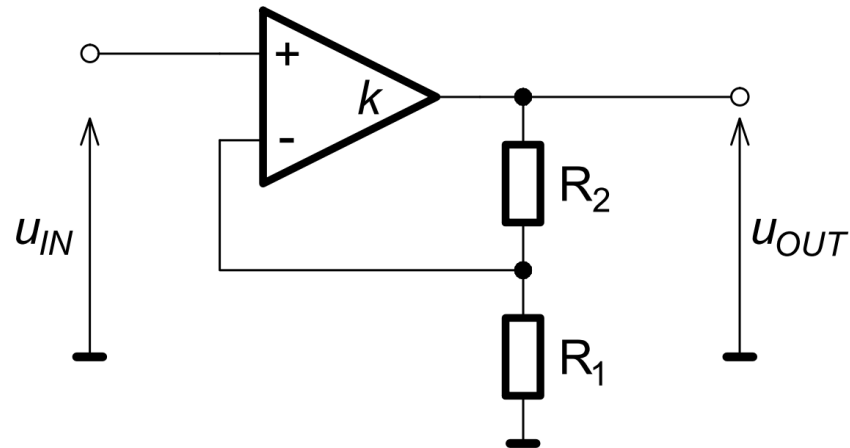
$$u_O = k(V_+ - V_-)$$

$$V_+ = u_{IN}$$

$$V_- = u_O \frac{R_1}{R_1 + R_2}$$

$$u_O = k \left(u_{IN} - u_O \frac{R_1}{R_1 + R_2} \right)$$

$$\frac{u_O}{u_{IN}} = \frac{k(R_1 + R_2)}{kR_1 + R_1 + R_2}$$



Introduction to Operational Amplifiers

BASIC AMPLIFIER: NON-INVERTING CONFIGURATION WITH INFINITE AMPLIFICATION $k \rightarrow \infty$

$$u_O = k u_D$$

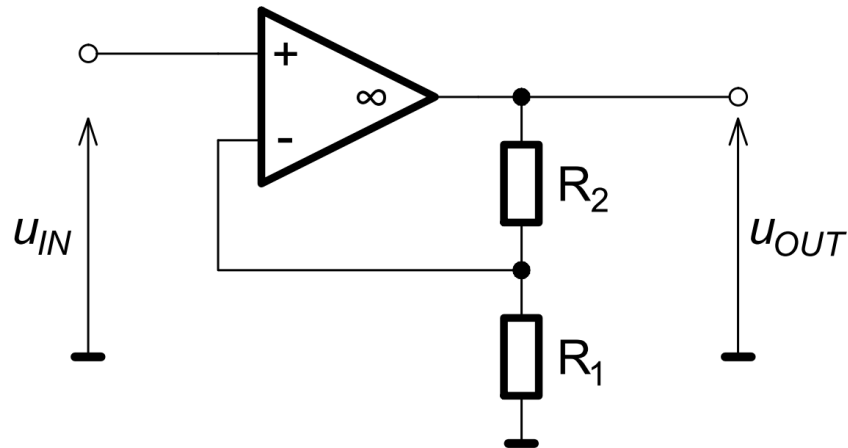
$$u_O = k(V_+ - V_-)$$

$$V_+ = u_{IN}$$

$$V_- = u_O \frac{R_1}{R_1 + R_2}$$

$$u_O = k \left(u_{IN} - u_O \frac{R_1}{R_1 + R_2} \right)$$

$$\frac{u_O}{u_{IN}} = \frac{k(R_1 + R_2)}{kR_1 + R_1 + R_2} \stackrel{k \rightarrow \infty}{=} 1 + \frac{R_2}{R_1}$$



Introduction to Operational Amplifiers

BASIC AMPLIFIER: INVERTING CONFIGURATION WITH FINITE AMPLIFICATION k

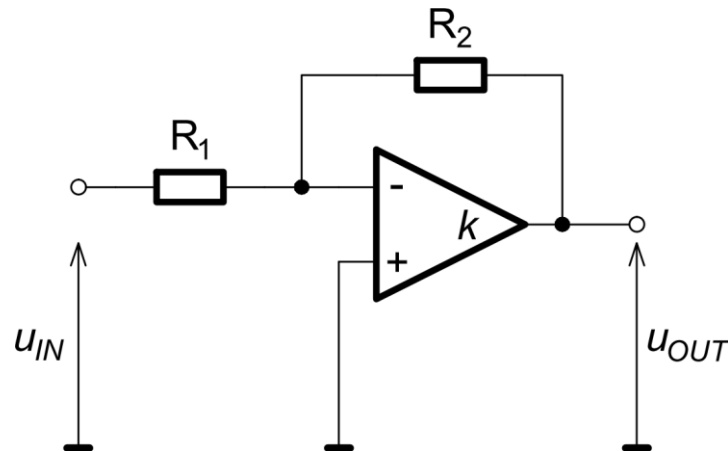
$$u_O = ku_D$$

$$u_O = k(V_+ - V_-)$$

$$V_+ = 0$$

$$V_- = ?$$

How to quickly find
the formula for V_- ?



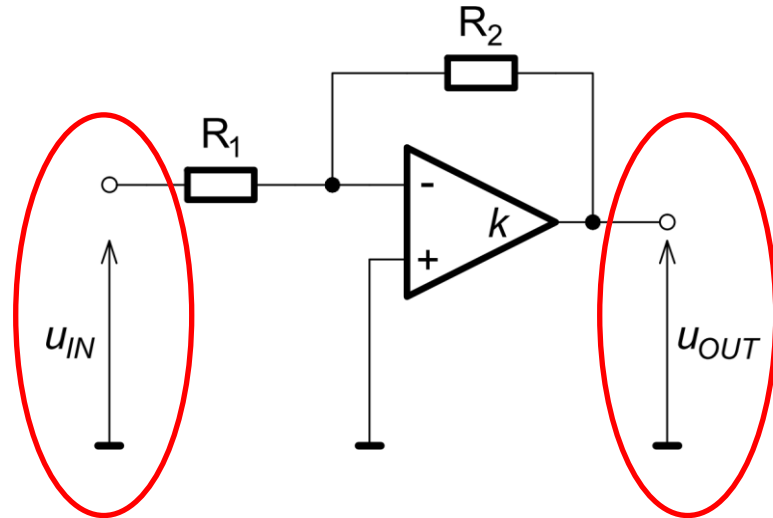
Introduction to Operational Amplifiers

BASIC AMPLIFIER: INVERTING CONFIGURATION WITH FINITE AMPLIFICATION k

*Determining V_- with the use of superposition principle
(for linear circuits only)*

Two sources are coincident
with V_- i.e., u_{IN} and u_{OUT}

$$V_- = V_-^{(u_{IN})} + V_-^{(u_{OUT})}$$



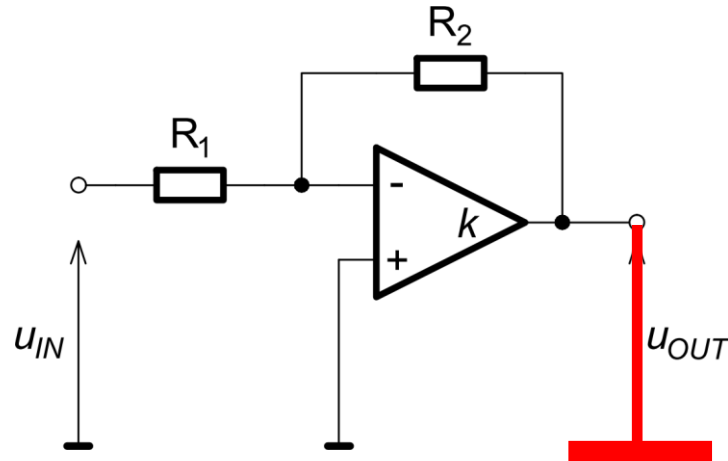
Introduction to Operational Amplifiers

BASIC AMPLIFIER: INVERTING CONFIGURATION WITH FINITE AMPLIFICATION k

*Determining V_- with the use of superposition principle
(for linear circuits only)*

Impact of u_{IN} on V_- :

$$V_-^{(u_{IN})} = u_{IN} \frac{R_2}{R_1 + R_2}$$



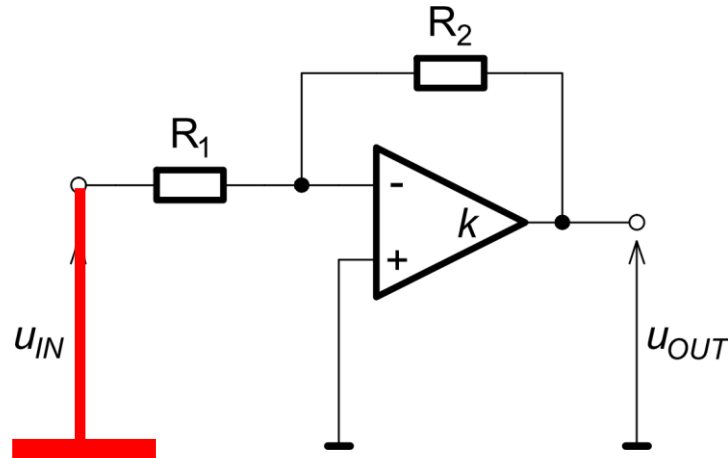
Introduction to Operational Amplifiers

BASIC AMPLIFIER: INVERTING CONFIGURATION WITH FINITE AMPLIFICATION k

*Determining V_- with the use of superposition principle
(for linear circuits only)*

Impact of u_{OUT} on V_- :

$$V_-^{(u_{OUT})} = u_{OUT} \frac{R_1}{R_1 + R_2}$$



Introduction to Operational Amplifiers

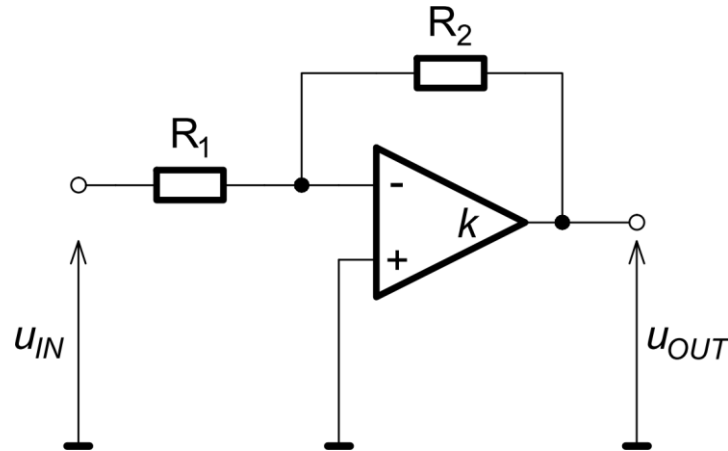
BASIC AMPLIFIER: INVERTING CONFIGURATION WITH FINITE AMPLIFICATION k

*Determining V_- with the use of superposition principle
(for linear circuits only)*

Expression V_- :

$$V_- = V_-^{(u_{IN})} + V_-^{(u_{OUT})}$$

$$V_- = u_{IN} \frac{R_2}{R_1 + R_2} + u_{OUT} \frac{R_1}{R_1 + R_2}$$



Introduction to Operational Amplifiers

BASIC AMPLIFIER: INVERTING CONFIGURATION WITH FINITE AMPLIFICATION k (continuation)

$$u_O = ku_D$$

$$u_O = k(V_+ - V_-)$$

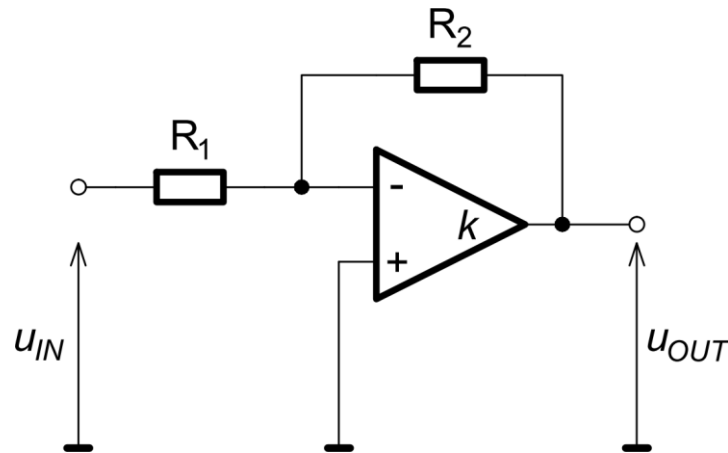
$$V_+ = 0$$

$$V_- = u_{IN} \frac{R_2}{R_1 + R_2} + u_{OUT} \frac{R_1}{R_1 + R_2}$$

$$u_{OUT} = -kV_- \Leftrightarrow V_- = -\frac{u_{OUT}}{k}$$

$$-\frac{u_{OUT}}{k} = u_{IN} \frac{R_2}{R_1 + R_2} + u_{OUT} \frac{R_1}{R_1 + R_2}$$

$$\frac{u_{OUT}}{u_{IN}} = -\frac{kR_2}{kR_1 + R_1 + R_2}$$



Introduction to Operational Amplifiers

BASIC AMPLIFIER: INVERTING CONFIGURATION WITH INFINITE AMPLIFICATION $k \rightarrow \infty$

$$u_O = ku_D$$

$$u_O = k(V_+ - V_-)$$

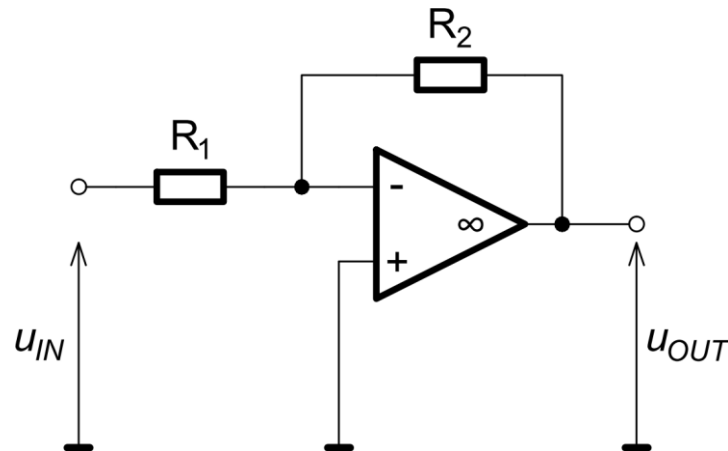
$$V_+ = 0$$

$$V_- = u_{IN} \frac{R_2}{R_1 + R_2} + u_{OUT} \frac{R_1}{R_1 + R_2}$$

$$u_{OUT} = -kV_- \Leftrightarrow V_- = -\frac{u_{OUT}}{k}$$

$$-\frac{u_{OUT}}{k} = u_{IN} \frac{R_2}{R_1 + R_2} + u_{OUT} \frac{R_1}{R_1 + R_2}$$

$$\frac{u_{OUT}}{u_{IN}} = -\frac{kR_2}{kR_1 + R_1 + R_2} \stackrel{k \rightarrow \infty}{=} -\frac{R_2}{R_1}$$



Introduction to Operational Amplifiers

FAST METHOD FOR DETERMINING AMPLIFICATION FOR OPERATIONAL AMPLIFIERS WITH INFINITE AMPLIFICATION $k \rightarrow \infty$

Step 1: determine V_+

$$V_+ = f(u_{IN}, u_{OUT}, R, L, C)$$

Step 2: determine V_-

$$V_- = f(u_{IN}, u_{OUT}, R, L, C)$$

Step 3: write:

$$V_+ = V_- \quad (\text{using the above expressions})$$

Why?

$$u_O = ku_D = k(V_+ - V_-)$$

$$\left. \begin{array}{l} |u_O| < \infty \\ k \rightarrow \infty \end{array} \right\} u_D \rightarrow 0 \Leftrightarrow V_+ = V_-$$

Step 4: rearrange the formula to obtain:

$$\frac{u_O}{u_{IN}} = f(R, L, C)$$

Introduction to Operational Amplifiers

1st FAST METHOD FOR DETERMINING AMPLIFICATION FOR OPERATIONAL AMPLIFIERS WITH INFINITE AMPLIFICATION $k \rightarrow \infty$ (example)

Step 1: determine V_+

$$V_+ = u_{IN2} \frac{R_4}{R_3 + R_4}$$

Step 2: determine V_-

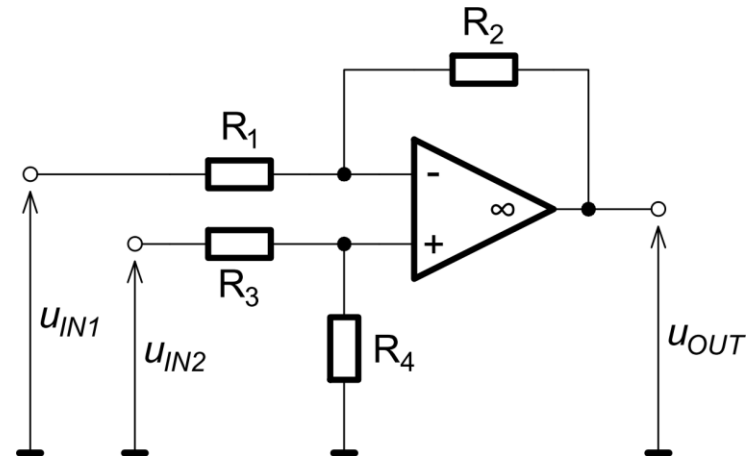
$$V_- = u_{IN1} \frac{R_2}{R_1 + R_2} + u_{OUT} \frac{R_1}{R_1 + R_2}$$

Step 3: write:

$$V_+ = V_-$$

$$u_{IN2} \frac{R_4}{R_3 + R_4} = u_{IN1} \frac{R_2}{R_1 + R_2} + u_{OUT} \frac{R_1}{R_1 + R_2}$$

$$u_{OUT} = u_{IN1} \frac{R_2}{R_1} - u_{IN2} \frac{R_4(R_1 + R_2)}{R_1(R_3 + R_4)}$$



Assuming $R_1 = R_2 = R_3 = R_4$

$$u_{OUT} = u_{IN1} - u_{IN2}$$

Introduction to Operational Amplifiers

2nd FAST METHOD FOR DETERMINING AMPLIFICATION FOR OPERATIONAL AMPLIFIERS WITH INFINITE AMPLIFICATION $k \rightarrow \infty$ (example)

Step 1: Find virtual ground:

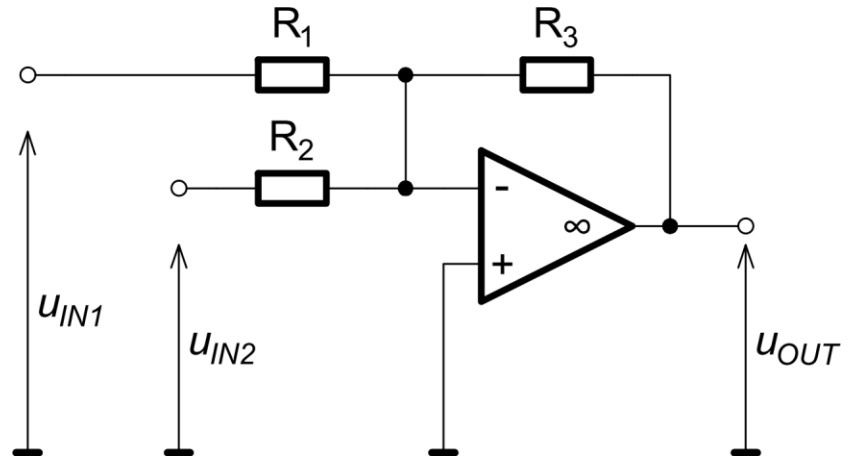
$$V_+ = 0 \Rightarrow V_- = 0$$

Step 2: calculate sum of currents for virtual ground:

$$\frac{u_{IN1}}{R_1} + \frac{u_{IN2}}{R_2} = -\frac{u_{OUT}}{R_3}$$

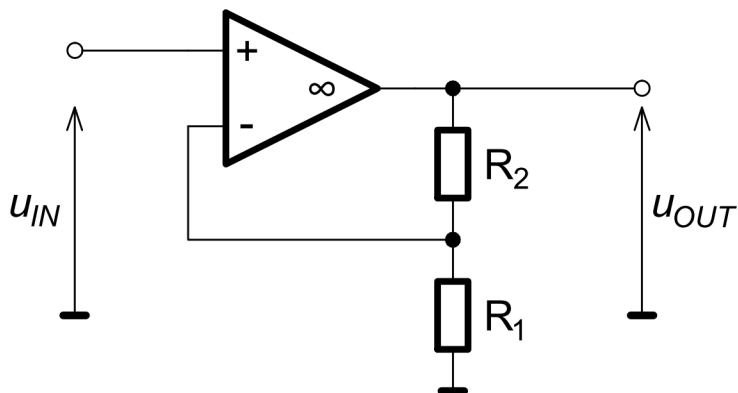
Step 3: rearrange the formula:

$$u_{OUT} = -\left(u_{IN1} \frac{R_3}{R_1} + u_{IN2} \frac{R_3}{R_2} \right)$$



Introduction to Operational Amplifiers

FEEDBACK ANALYSIS

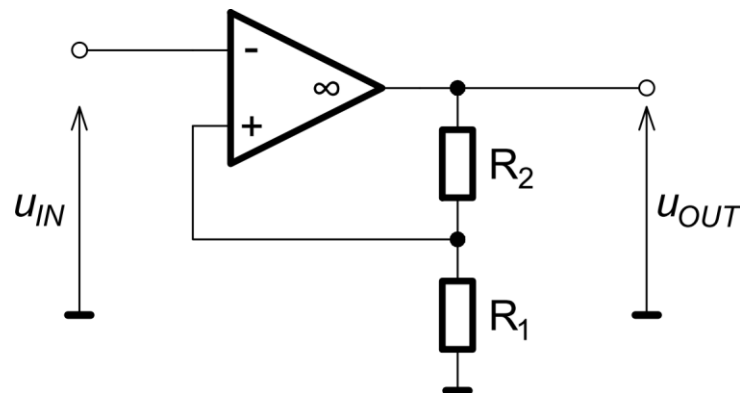


$V_+ \uparrow \Rightarrow u_D \uparrow \Rightarrow u_{OUT} \uparrow \Rightarrow V_- \uparrow \Rightarrow u_D \downarrow$

Negative feedback

Negative feedback suppresses amplification of the operational amplifier: the entire circuit has lower amplification (but stable)

LINEAR CIRCUIT



$V_- \uparrow \Rightarrow |u_D| \uparrow \Rightarrow |u_{OUT}| \uparrow \Rightarrow |V_+| \uparrow \Rightarrow |u_D| \uparrow \uparrow$

Positive feedback

Positive feedback magnifies amplification of the operational amplifier: any deviation of u_D from 0 saturates operational amplifier

$(u_{OUT} \rightarrow \pm V_{CC})$

NONLINEAR CIRCUIT

Introduction to dB scale

Amplification/attenuation in linear scale exhibits very wide range e.g.:

$$\frac{u_O}{u_{IN}} \approx 1 \quad \text{e.g., for buffers}$$

$$\frac{u_O}{u_{IN}} \approx 10^6 \quad \text{e.g., for operational amplifiers}$$

$$\frac{u_O}{u_{IN}} \approx \frac{1}{10^6} \quad \text{e.g., for filters (stop-band)}$$

To suppress the range of amplification/attenuation values and ease the calculation, **decibels (dB)** are introduced to express power ratio:

$$P_1^{dB} - P_2^{dB} = 10 \log_{10} \left(\frac{P_1^{lin}}{P_2^{lin}} \right)$$

Introduction to dB scale

Decibels can also be used to express absolute power values:

$$P^{dB} = 10 \log_{10} \left(\frac{P^{lin}}{1[W]} \right)$$

Absolute power expressed in dB; reference to 1W

$$P^{dBm} = 10 \log_{10} \left(\frac{P^{lin}}{1[mW]} \right)$$

Absolute power expressed in dBm (millidecibels); reference to 1mW

$$P^{dBm} = P^{dB} + 30$$

Inverse calculations:

$$P^{lin} = 10^{\frac{P^{dBm}}{10}} [mW] \quad P^{lin} = 10^{\frac{P^{dB}}{10}} [W]$$

Introduction to dB scale

Decibels can also be calculated with the use of voltages and currents:

$$P_1^{dB} - P_2^{dB} = 20 \log_{10} \left(\frac{U_1^{lin}}{U_2^{lin}} \right)$$

$$P_1^{dB} - P_2^{dB} = 20 \log_{10} \left(\frac{I_1^{lin}}{I_2^{lin}} \right)$$

since

$$P \approx U^2 \quad P \approx I^2$$

Introduction to dB scale

How to calculate linear value from the one given in dB scale?
(with resolution of 1 dB)

$$10\log_{10}\left(\frac{2P_1^{lin}}{P_1^{lin}}\right) = 3.0103 \approx 3$$

$$10\log_{10}\left(\frac{10P_1^{lin}}{P_1^{lin}}\right) = 10$$

Any integer P can be expressed as a superposition of even number (e.g., 10) and odd number (e.g., 3)

$$P \in \mathbb{Z} \Leftrightarrow P = a \cdot 3 + b \cdot 10$$

where

$$a, b \in \mathbb{Z}$$

Introduction to dB scale

How to calculate linear value from the one given in dB scale?
(with resolution of 1 dB)

By application of the following basic rules

$$\log_{10}(ab) = \log_{10}(a) + \log_{10}(b)$$

$$\log_{10}(a^c) = c \log_{10}(a)$$

We can write

$$P^{lin} = 2^a \cdot 10^b [W]$$

Introduction to dB scale

How to calculate linear value from the one given in dB scale?
(with resolution of 1 dB)

Example: $P = 14$ dB. What is the corresponding values in linear scale?

$$14 = \boxed{-2} \cdot \boxed{3} + \boxed{2} \cdot \boxed{10}$$
$$P^{lin} = \boxed{2}^{\boxed{-2}} \cdot \boxed{10}^{\boxed{2}} = \frac{10^2}{2^2} = 25[W]$$

From exact calculations:

$$P^{lin} = 10^{\frac{14}{10}} = 25.1189[W]$$

Calculation error is equal to -0.1189 W